



DURBAN UNIVERSITY OF TECHNOLOGY
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FACULTY OF ACCOUNTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY

2024
FEBRUARY SPECIAL EXAMINATION

INSTRUCTIONAL PROGRAMME : B INFORMATION & COMMUNICATION TECHNOLOGY
INSTRUCTIONAL OFFERING : INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

PAPER NUMBER : 1

SUBJECT CODE/S : IPRT301

DATE : 29 FEBRUARY 2024

DURATION : 3 HOURS

TIME : 09:00-12:00

TOTAL MARKS : 100

NUMBER OF PAGES : 6 (INCLUDING COVER PAGE)

EXAMINER/S : BOAMAH-ABU, CHARLES (PhD)

MODERATOR/S : IBIDUN OBAGBUWA, CHRISTIANA (PhD)

INSTRUCTIONS/REQUIREMENTS:-

1. This examination is closed book
2. Answer all questions
3. Where stated "briefly", please answer in a few lines
4. Ensure your handwriting is readable

Do not turn the page until permission is given

Question 1

10 marks

Indicate whether each of following statement is **true** or **false**.

1. Integrative programming allows software developers to create platform-independent applications.
2. A design pattern suggests the programming language to solve a problem.
3. A design pattern is only a guideline on how to solve a problem.
4. Any class can be declared an abstract class.
5. An abstract class cannot extend another abstract class.
6. Method override occurs in the same class.
7. An XML document is encoded and decoded in data exchange among different platforms.
8. CORBA supports the implementation of server and client applications in different programming languages.
9. JSP code contains a main method where execution begins.
10. JSP code runs in a client browser.

Question 2

12 marks

Briefly answer the following questions

1. Explain why it is preferable to design interfaces or abstract classes, rather than concrete classes. (2 marks)
2. Describe the **Singleton** design pattern. Illustrate with an example where this pattern can be used. (4 marks)
3. Give one reason why server-side scripts are preferable to client-side scripts. (2 marks)
4. Explain why the following are important factor to consider in distributed applications. (4 marks)
 - a. Exception handling
 - b. Multithreading.

Question 3

20 marks

The code below is part of an XML document (**marksheet.xml**). Each **Student** node contains **id** (a unique identifier), **name** (of student), **dp** mark and **exam** mark.

```
<?xml version="1.0" encoding="UTF-8"?>
<Marksheet>
  <Student id="20220001">
    <name>J May</name>
    <dp>40</dp>
    <exam>40</exam>
  </Student>
  <Student id="20220002">
    <name>K Lo</name>
    <dp>50</dp>
    <exam>50</exam>
  </Student>
  ....
</Marksheet>
```

1. Complete the DTD file (**marksheet.dtd**), below, which defines the structure in **marksheet.xml**. (7 marks)

```
<?xml version='1.0' encoding='UTF-8'?>
<!ELEMENT Marksheet (Student)*>
```

2. Complete the tag below which links **marksheet.xml** to the structure in **marksheet.dtd**. (3 marks)

```
<!DOCTYPE _____>
```

3. The data in **marksheet.xml** is to be displayed in a web page, as shown below.

http://localhost:8080/specialexamweb/marksheet.xml

Final marks

Id	Name	Dp	Exam	Final mark
20220001	J May	40	40	40
20220002	K Lo	50	50	50
20220004	Awo Yaa	70	80	76

Assume the final mark is calculated by adding 40% of **dp** mark and 60% of **exam** mark, i.e., $((0.4*dp)+(0.6*exam))$. Complete the parts of the XSL document (**marksheet.xsl**), below, to display the details of each student in the web page. (8 marks)

```
....
<xsl:template match="/">
....
  <xsl:for-each select=_____>
    <xsl:_____ /> </td>
    <xsl:_____ /> </td>
    <xsl:_____ /> </td>
    <xsl:_____ /> </td>
    <xsl:_____ /> </td>
  </xsl:for-each>
....
```

4. Complete the tag, below, to link **marksheet.xml** to the format in **marksheet.xsl**. (2 marks)

```
<?xml-stylesheet type="_____" href="_____"?>
```

Question 4

13 marks

1. The underlined part of the code segment below shows interface IA implements interface IB. Indicate whether this is **valid** or **invalid**. Explain your answer. (2 marks)

```
public interface IA implements IB {
....
}
```

2. The underlined part of the code segment below shows interface IX extends IY. Indicate whether this is **valid** or **invalid**. Explain your answer. (2 marks)

```
public interface IX extends IY {
....
}
```

3. If class A is a subclass of class B and A implements the interfaces IM and IN, complete the class header below. (3 marks)

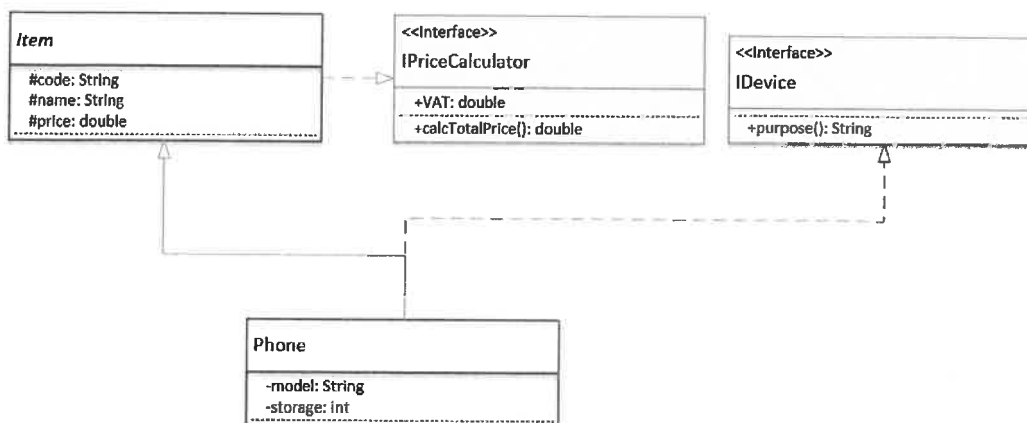
```
public class _____ {
....
}
```

4. Class C is an abstract class and a super class of a class D. Indicate whether each of the following statements is **valid** or **invalid**. Explain your answer. (6 marks)
- C c1=new C();
 - D d1=new D();
 - C c2=new D();

Question 5

10 marks

The UML diagram below shows **Item** contains **code**, **name** and **price** of an item. **Phone** contains the **model** and **storage** capacity. **IPriceCalculator** contains **VAT** (a constant, 15%) and **calcTotalPrice()**, which returns the total price as the total of price and VAT payable. The VAT payable is calculated as 15% of the price of the item. **IDevice** contains **purpose()**, which returns a string.



- Write the method **calcTotalPrice()** in **IPriceCalculator**. (2 marks)
- Write the implementation of **calcTotalPrice()** in **Item**. (2 marks)
- Write the class header for **Phone**. (2 marks)

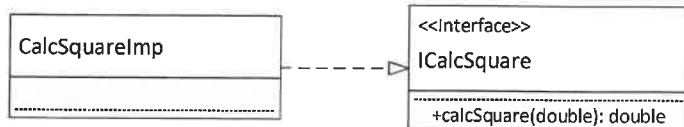
4. Using the classes **Item** and **Phone**, briefly explain the term “polymorphism”. (2 marks)
5. State whether the statement below is **valid** or **invalid**. Explain your answer. (2 marks)

`IPriceCalculator ip=new IPriceCalculator ();`

Question 6

25 marks

You are required to create an RMI application. The figure below shows **ICalcSquare** (the remote interface). The class **CalcSquareImp** implements **ICalcSquare**. The method **calcSquare(double)** takes a number as parameter and returns the square of the number.



1. Write the header of the remote interface **ICalcSquare**. (2 marks)
2. Write the class header for **CalcSquareImp**. (3 marks)
3. Write the method **calcSquare(double num)** in **CalcSquareImp**. (3 marks)
4. Complete the class **Server** (below) which creates, registers and binds an instance (**obj**) of the remote object. Bind the name **square** to the remote object; use port **1888**. (8 marks)

```

public class Server {
    public static void main(String[] args) _____ {
        _____
        _____
        _____
    }
}
  
```

5. Complete the class **Client** (below) which calls **calcSquare(double num)** of the remote object with the value 80.0 and displays the result on the screen. Use the local host. A typical output is shown below. (7 marks)

The square of 10.0 is 100.0

```

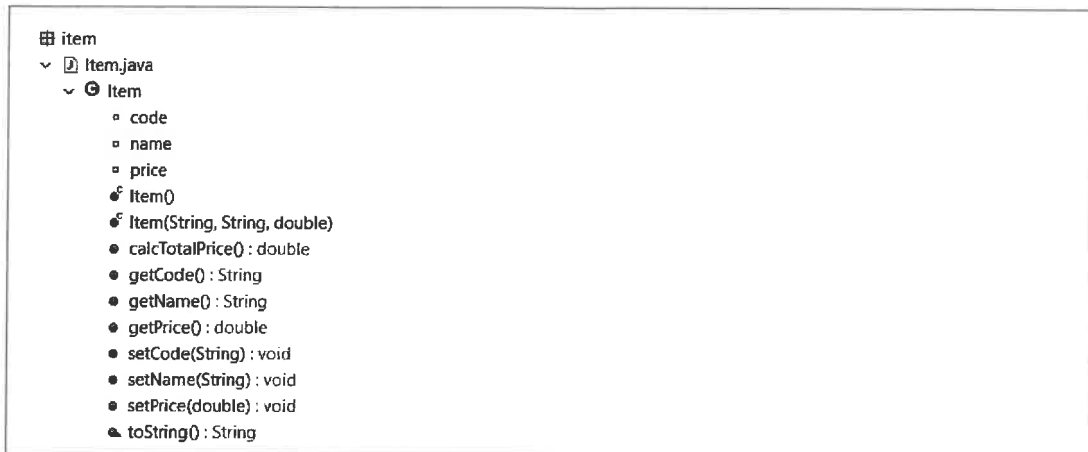
public class Client{
    public static void main(String[] args) _____{
        _____
        _____
    }
}
  
```

6. Briefly explain what will happen if the Client program in this RMI application is started before the Server program. (2marks)

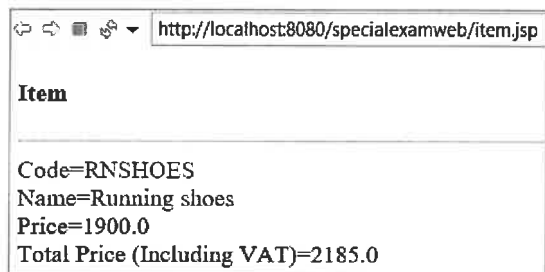
Question 7

10 marks

The package **item** (below) contains the class **Item** which contains **code**, **name** and **price** of an item. The method **calcTotalPrice()** returns the total price (i.e., price + VAT on an item). The method **toString()** returns the details (i.e., **code**, **name**, **price** and **total price**) of an item as a string.



The package **item** is imported into JSP code to display the details an item in a web page (as shown below).



Assume the code for the class **Item** has been implemented.

1. Complete the JSP page directive (below), which specifies the programming language and imports the class **Item**. (5 marks)

<%@ _____ %>

2. Write the JSP **scriptlet**, which creates an instance of **Item** and calls the appropriate method to display the details as shown in the web page. (5 marks)